

Singapore Institute of Technology
BEng (Hons) Telematics
(Intelligent Transportation System Engineering)
TLM2008 Internet Programming

Web Application Project

TRaffic INformation Display System(TRINDS)

Group Name: CARVID-6

Written By:	Neo Li Qin	(1801758)
	Ng Jun Hong	(1801289)
	Jeremy Lim	(1800960)
	Sean Sin Kiat Wei	(1802859)
	Khatcharin Ng	(1802186)
	Rycca Ng	(1801684)

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Objective of TRINDS

Our team is creating a web application that allows registered users to keep track of all the traffic news that is happening in Singapore. Users are allowed to submit any traffic news that is happening on the road. It allows users to report any traffic update via submitting an entry in our webpage. Entries will be saved to the database and reflected in our Statistics' page on our webpage.

The target viewers are users on the road who require the latest traffic updates so that they are able to plan their route for the journey beforehand.

The webpage application only allows registered users and admin to use all the functions available on our webpage. Only registered users will be able to submit an entry to report all types of traffic news such as accidents, vehicle breakdown, road works, road closures and heavy traffic. Users are able to view the latest traffic news using the interactive map where the different icons represent the type of traffic updates respectively.

Problem and Design Concept of TRINDS

Most web applications only allow users to view the road traffic that occurs on the road. However, there are still further improvements that can be made on the road. For example, the current web application does not have the function of submitting an entry of traffic update. It only waits for organisations such as the Land Transport Authority (LTA) to get the update before displaying the latest traffic news in their webpage. This might affect those drivers that are travelling towards the affected roads without getting the latest traffic updates.

Therefore, our team proposed the idea where users do not necessarily have to wait for the updates from the authorities. Instead, users are able to submit an entry for the latest traffic updates themselves or view the traffic updates made by other users. This allows all the users to be able to view the latest traffic updates. However, the admin needs to process the new entries submitted by the user and updates the interactive map.

Technologies used for TRINDS

For the website, our team used Moderna Bootstrap as our base bootstrap template to start the project and edited the codings using HTML, CSS, PHP script programming to better suit our project objectives. Mapbox was used for the users to interact through selecting the point of location on the Singapore map to indicate the issue when they are submitting a new traffic news entry. Google Maps API was used to display the location of previous issue entries, comments and types of issues submitted by users. As both the interactive maps have to be linked to both the user interface and database, PHP database connection and Javascript were used.

The design of the website design and the technologies that we used are able to address the problem that we have raised. Our users are able to view the latest traffic news from the map and are also able to submit new entries on the traffic news to update their fellow road users.

Demonstration script of TRINDS

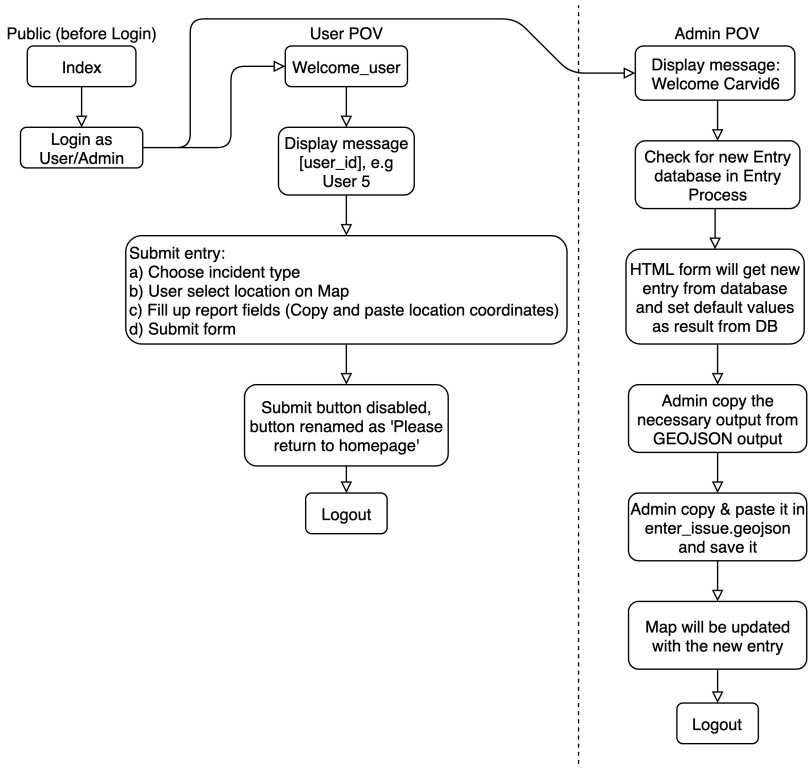


Figure 1: System flow of TRINDS

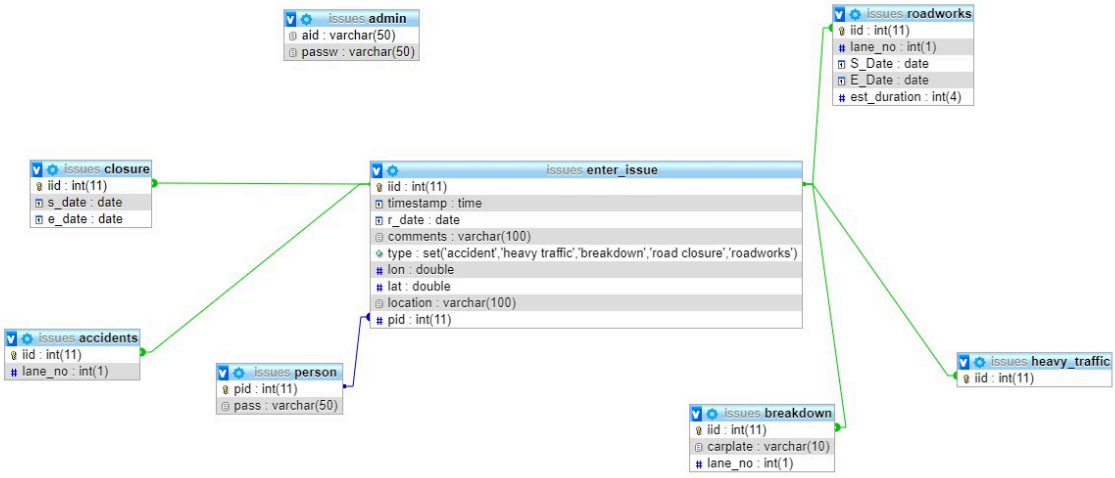


Figure 2: ER Diagram for overall program

Code implementation examples

```
<!-- ======= Header ======= -->
<header id="header" class="fixed-top ">
      <div class="container">

        <div class="logo float-left">
        <a href="index.php"><img src="assets/img/logo.png" alt="logo"></a>      <!-- TRINDS logo-->
        <!--<h1 class="text-light"><a href="index.php"><span>TRINDS</span></a></h1> !-->
        </div>

          <nav class="nav-menu float-right d-none d-lg-block"> <ul>
            <li><a href="index.php">Home</a></li>
            <li><a href="login_user.php">About Us</a></li>
            <li><a href="login_user.php">Map</a></li>
            <li><a href="login_user.php">Statistics</a></li>
            <li><a href="login_user.php">Contact Us</a></li>
            <li class="drop-down"><a href="#">Login</a>
                <ul>
                    <li><a href="login.php">Login as admin</a></li>
                    <li><a href="login_user.php">Login as user</a></li>
                </ul>
            </li>

          </nav><!-- .nav-menu -->

        </div>
 </header><!-- End Header -->
```

Figure 3: Header implementation for main index.php

```
<?php include 'header3.inc'?>
   <section id="loginsection">
   <form id="login"  method="post" action="check_login_user.php">
   <h3>User Login</h3>
   <b>LoginID</b><br><input type="text" name="id" id="id" size="30" style="height:20px;" ><br><br>
   <b>Password</b><br><input type="text" name="passwd" id="passwd" size="30" style="height:20px;"/><br><br><br><br>
    <input type="submit" name="submit" value="Login"/>
    <p>Forgotten your password?</p>
    <?php if(isset($_GET['error'])) echo "<b>Invalid username or password</b>";?>
   </form>
   </section>
```

Figure 4: PHP contact form implementation using POST method in login_user.php

```
<div class="col-lg-6">
    <label>Please select the issue to report:</label><br>
    <li><a href="entry_A.php">Accident</a></li>
    <li><a href="entry_H.php">Heavy traffic</a></li>
    <li><a href="entry_B.php">Breakdown</a></li>
    <li><a href="entry_C.php">Road closure</a></li>
    <li><a href="entry_W.php">Road works</a></li>
</div>
```

Figure 5: User entry submission for type of incident in entry.php

```
<!-- Set map -->
<div id="map"></div>
<div id="displayhelp">
    Select the location you want to report an incident:
</div>
```

Figure 6: Display Mapbox in entryA.php

```
<!-- Mapbox Script -->
<script>
  mapboxgl.accessToken = 'pk.eyJ1IjoiZnVndWVtYXN0ZXIiLCJhIjoiY2s3aG45dDQ0MGM4YzNmbGxiM3JqaXBlaCJ9.UXkXKi2Cq8veJTtkUYq8wg';

      // Set scorlling boundaries to Singapore Map
      var bounds = [
          [103.56480861951604, 1.154021455003118], // Southwest coordinates[lng, lat]
          [104.13164125586022, 1.518896810581026]  // Northeast coordinates[lng, lat]
      ];

      // Set map
      var map = new mapboxgl.Map({
          container: 'map', // container id
          style: 'mapbox://styles/fuguemaster/ck7ho8svq54td1ip8d8w18vmt', // stylesheet location
          center: [103.8198, 1.3521], // starting position [lng, lat], Singapore
          zoom: 12, // starting zoom
          maxBounds: bounds // Sets bounds as max
      });

      // Marker settings
          var marker = new mapboxgl.Marker({
          draggable: false
          });
      // Create marker: mouse on click event
      map.on('click', function(e) {

          marker.setLngLat(e.lngLat);
          marker.addTo(map);

          document.getElementById('displayhelp').innerHTML =
                  'Select the location you want to report an incident: '
                  + '<br>' + 'Longitude: ' + e.lngLat.lng + '<br>' + 'Latitude: ' + e.lngLat.lat;

      });
```

Figure 7: Mapbox Script that sets the map settings and functions for user interaction

```
<h2>Accident issue form</h2>
<form action="confirmationA.php" method="post" class="php-email-form" onsubmit="return checkForm(this);">

  <div class="form-row">
    <div class="col-md-6 form-group">
      <input type="text" name="name" class="form-control" id="name" placeholder="Your Name" data-rule="minlen:4" data-msg="Please enter at least 4 chars" />
      <div class="validate"></div>
    </div>
    <div class="col-md-6 form-group">
      <input type="email" class="form-control" name="email" id="email" placeholder="Your Email" data-rule="email" data-msg="Please enter a valid email" />
      <div class="validate"></div>
    </div>
    <div class="col-md-6 form-group">
        <input type="text" name="pid" class="form-control" id="pid" placeholder="Your PID" data-rule="minlen:1" data-msg="Please enter <b>your</b> PID." />
      <div class="validate"></div>
    </div>
  </div>

  <div class="form-row">
    <div class="col-md-6 form-group">
      <input type="text" name="lon" class="form-control" id="lon" placeholder="Your Longitude" data-rule="minlen:6" data-msg="Please enter at least 6 chars" />
      <div class="validate"></div>
    </div>
    <div class="col-md-6 form-group">
      <input type="text" name="lat" class="form-control" id="lat" placeholder="Your Latitude" data-rule="minlen:4" data-msg="Please enter at least 4 chars" />
      <div class="validate"></div>
    </div>
  </div>

  <div>
    <input type="text" name="location" class="form-control" id="location" placeholder="Your location based on nearest road name">
  </div>

    <div class="form-group">
      <textarea class="form-control" name="message" rows="5" data-rule="required" data-msg="Please write something for us" placeholder="Message"></textarea>
      <div class="validate"></div>
    </div>

    <div class="form-group">
                    <label>Select lane number applicable for <b>accident</b> report</label>
          <select class="c-form-type" id="rLane_no" min="0" data-rule="required" data-msg="Please select lane">
              <option value="Type">Select...</option>
              <option value="nil">-</option>
              <option value="1">1</option>
              <option value="2">2</option>
              <option value="3">3</option>
              <option value="4">4</option>
              <option value="5">5</option>
          </select><br>
        <div class="validate"></div>
    </div>
    <div class="mb-3">
      <div class="loading">Loading</div>
      <div class="error-message"></div>
      <div class="sent-message">Your message has been sent. Thank you!</div>
    </div>
    <div class="text-center"><input type="submit" value="Submit" name="submitBtn"></div>

  </form>
```

Figure 8: User entry submission for reporting of an accident in entryA.php

```
    //get variable from form input for accident and validate
    $aLane_no = filter_input(INPUT_POST, 'lane_no');
    $aMessage = filter_input(INPUT_POST,'message');
    $aLon = filter_input(INPUT_POST,'lon');
    $aLat = filter_input(INPUT_POST, 'lat');
    $aLocation = filter_input(INPUT_POST, 'location');
    $aPid = filter_input(INPUT_POST, 'pid');
```

```
$conn1->query("INSERT INTO enter_issue(timestamp, r_date, comments, type, lon, lat, location, pid, icon)"
    . "VALUES (NOW(), NOW(), '$aMessage', 'accident', '$aLon', '$aLat', '$aLocation', '$aPid', 'assets/geojson/accident.jpg');");

$iid=$conn1->insert_id;
$conn2->query("INSERT INTO accidents(iid,lane_no) VALUES ($iid,'$aLane_no');");
```

Figure 9: Information from contact form parsed to database in confirmationA.php

```
// Generate GeoJSON from contact form
const createGeoJSON = () => {
    const data = [
        {
            lat: document.getElementById("latitude").value,
            lng: document.getElementById("longitude").value
        }
    ];
    data[0]["iid"] = document.getElementById("iid_value").value;
    data[0]["timestamp"] = document.getElementById("timestamp_value").value;
    data[0]["comments"] = document.getElementById("comments_value").value;
    data[0]["type"] = document.getElementById("type_value").value;
    data[0]["location"] = document.getElementById("location_value").value;
    data[0]["pid"] = document.getElementById("pid_value").value;
    data[0]["icon"] = document.getElementById("icon_value").value;




    const dataGeoJSON = GeoJSON.parse(data, { Point: ["lat", "lng"] });
    out = JSON.stringify(dataGeoJSON, null, 4);
    document.getElementById("output").value = out;
}
createGeoJSON();
```

Figure 10: Generate GeoJSON from contact form in adminEntryProcessing.php


```
{
  "type": "FeatureCollection",
  "features": [
  {
    "type": "Feature",
    "geometry": {
       "type": "Point",
       "coordinates":  [ 103.763696,1.312923 ]
    },
    "properties": {
    "iid":1,
    "timestamp":"5:45",
    "comments":"",
    "type":"Accident",
    "location":"Clementi Ave 3",
    "pid":1,
    "icon":"assets/geojson/accident.jpg"
    }
  },
```


Figure 11: GeoJSON file which stores the new entry from user in enter_issue.geojson

```
var map;
var geojson;

function initMap() {  // initialise map

  map = new google.maps.Map(document.getElementById('map'), {
    center: {lat: 1.3521, lng: 103.8198},
    zoom: 12
  });

  map.data.setStyle(function(feature) {
    return {icon:feature.getProperty('icon')};
  });


  //Load GeoJson file
  geojson = map.data.loadGeoJson(
      'assets/geojson/enter_issue.geojson');


  // Show info box on mouse click event
  map.data.addListener('click', function(event) {
    var html = 'Issue no.: '+event.feature.getProperty('iid')+'<br>'
             + 'Location: '+event.feature.getProperty('location')+'<br>'
             + 'Type: '+event.feature.getProperty('type')+'<br>'
             + 'Time reported: '+event.feature.getProperty('timestamp')+'<br>'
             + 'Comments: '+event.feature.getProperty('comments');
    document.getElementById('info-box').innerHTML = html;
  });

}
</script>


<script src="https://maps.googleapis.com/maps/api/js?key=AIzaSyBf8sdRfN_rf0iZCeAZ5gCTSBkmHlCbmVE&callback=initMap"
async defer></script>
```

Figure 12: Interactive map display utilising the enter_issues.geojson file and Google Maps API in map_user.php

Improvements to be made on TRINDS

Compared to the current contact form for our website design, TRINDS' entry form can be made more user friendly through a nested dropdown list form, where the selection of different types of issues will display the fields of entries required to be filled. The entry form can also be able to redirect the users back to the homepage upon their submission with a notification message popup to inform them on the success.

The most inconvenient point for the users is to have to copy and paste the location coordinates to fill up the form. We can improve them by finding ways to have the entry autofilled when the user selects a location on the interactive map. We can also cut the width of the drop down menu for the navigation bar so that no 'glitches' will be shown when we hover the mouse from one option to the next option.

In addition, we can create a registration page for new users to create a new account into the database. For the statistics on the overall incidents, we can design a pie chart to show for the type of incidents which have a higher percentage of occurance compared to other types. The area of which incidents happen the most can also be included.

Work distribution to create TRINDS

Name	Work distribution
Neo Li Qin	Overall Javascript programming, using git merge to combine all team members' codes into one project folder, Mapbox integration
Ng Jun Hong	Homepage, Header, Footer, Logo design, Bootstrap template manipulation
Rycca Ng	Header, login pages, Overall PHP files management and design
Jeremy Lim	Google map integration, PHP files for all statistics display, Report writing
Sean Sin	Database manipulation, Confirmation page, Project video editing
Khatcharin Ng	Entry form/pages, Confirmation pages, Report writing

For more information for work distribution, can find out more in our Github: CARvid6

Group Reflections

During this project, the team made use of Github to manage our shared project repository and used GitExtensions to monitor everyone's progress and contribution. The team had managed to learn Git and apply it into our project management. It boosts our efficiency in terms of troubleshooting for code differences and combining everyone's compiled codes into one project.

We have learnt how to apply the skills that we learnt from the lectures into the project. In order to do this project, we were required to revise what we were taught in Database and Information System (TLM2004) during the last trimester. We had to link our entry form submission to the database before sending the results back to the website to display on the map and statistics.

Most of our takeaways from this project are the knowledge of HTML, CSS, PHP and Javascript. As most of us are not familiar on how to approach the project, we relied on external sources such as w3school to help us get started and bootstrap templates to get an overall idea of how we could design our webpage. In order to make this web application, we went through tons of troubleshooting, problem solving and error compiling of the source codes to get to the final completed webpage.

Overall, it has been a good exposure working on the project and brushing up on our programming skills such as HTML, PHP, CSS and Javascript. We also had a deeper understanding on how the user interface works with the database to both input new data and display the updated database data onto the map and statistics page.

References

Free Bootstrap template for corporate - Moderna. (2020, February 4).
Retrieved March 3, 2020, from
https://bootstrapmade.com/free-bootstrap-template-corporate-moderna/

HTML. (n.d.).
Retrieved March 4, 2020, from https://www.w3schools.com/

Mapbox. (n.d.).
Retrieved March 10, 2020, from
https://www.mapbox.com/?utm_medium=sem&utm_source=google&utm_campaign=sem|google|brand|chko-googlesearch-pr01-mapboxbrand-br.exact-intl-landingpage-search&utm_term=brand&utm_content=chko-googlesearch-pr01-mapboxbrand-br.exact-intl-landingpage-search&gclid=CjwKCAiA44LzBRB-EiwA-jJipNywqEJr0aZDJvH55m-zks8tHtEVMZ_PMTqLWroPB9lGlGpuEMP5pBoCiXQQAvD_BwE

Google Cloud. (n.d.).
Retrieved March 11, 2020, from https://cloud.google.com/

Flaticon, the largest database of free vector icons. (n.d.).
Retrieved March 23, 2020, from https://www.flaticon.com/

Raboy, N. (2019, May 21). Format Data into GeoJSON with JavaScript to be used with HERE XYZ - HERE Developer.
Retrieved March 20, 2020, from
https://developer.here.com/blog/format-data-into-geojson-with-javascript-to-be-used-with-here-xyz

Credits

Template Name: Moderna - v2.0.0
Template URL: https://bootstrapmade.com/free-bootstrap-template-corporate-moderna/
Author: BootstrapMade.com
License: https://bootstrapmade.com/license/
-Credits are also included in our source codes(header and footer).

LogoMakr- https://logomakr.com/
-Credits are also included in our source codes(footer).